Department of Computer Science and Engineering
BRAC University
CSE 260: Digital Logic Design

Experiment # 6: Design circuit using encoder & decoder.

Objective:

- To get familiarized with ***74ls138 [decoder]; 74ls148 [encoder]***
- To gain experience working with practical circuits.

Required Components:

1. IC 74138
2. IC 74148
3. IC 7404

Theory:

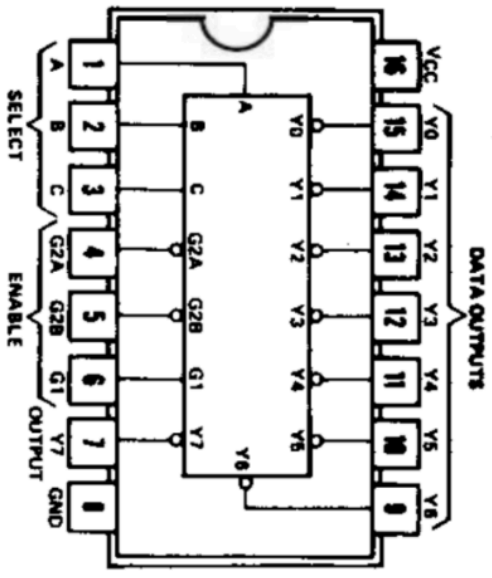
- **Enable Pin:** An enable pin is a special input in a digital circuit that works like an ON/OFF switch. It controls whether the circuit or a part of it is active or inactive.
- **Active Low Circuit:** Active-low means the circuit turns on or performs its function when the input is 0 (low voltage). For example, if an **enable pin** is **active-low**, it means you must give 0 (low voltage/ GND) to turn the circuit ON.

Task :

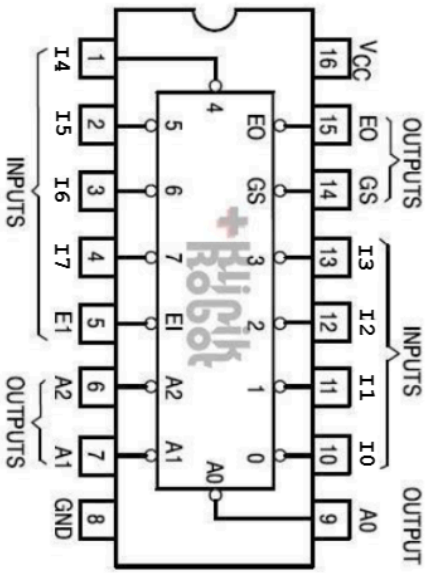
Design a circuit using 3-to-8 decoder, 8-to-3 encoder and a not gate to implement the function Y = (X + 2) mod 8, where X is a 3-bit binary input. The final output must be active high.

Sample Input-Output:

Input	Output	Explanation
001	011	$001_2 + 010_2 = 011_2$; $011_2 = 3_{10}$; 3 mod 8 = 3. So, the final output is 011_2
111	001	$111_2 + 010_2 = 1001_2$; $1001_2 = 9_{10}$; 9 mod 8 = 1. So, the final output is 001_2



74138



74148

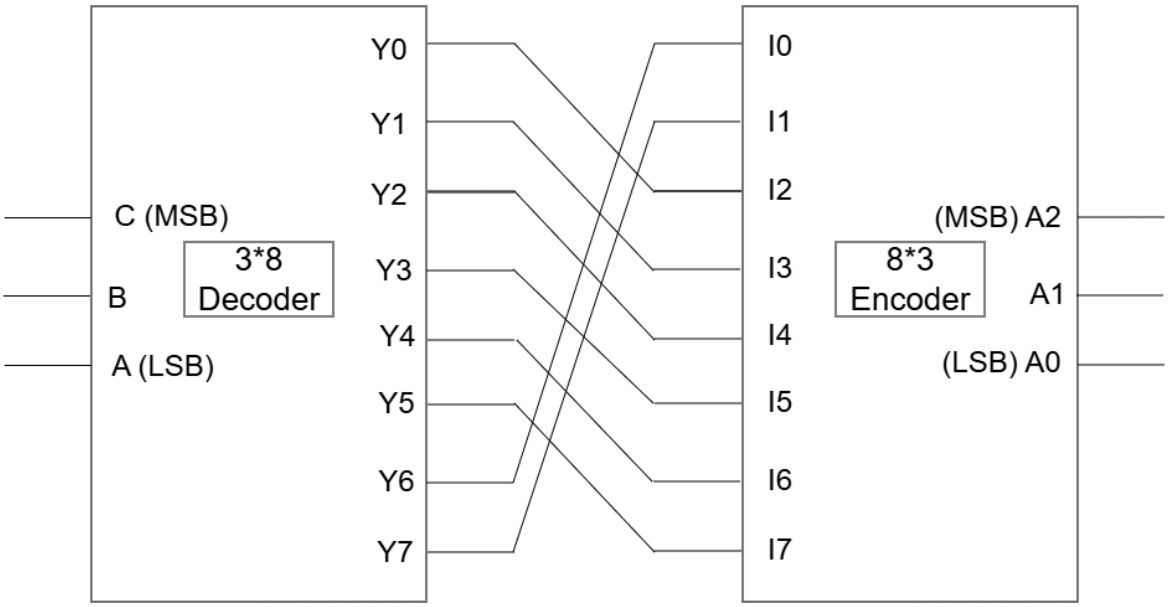
Procedure:

We will design the circuit using a 3:8 Decoder, 8:3 Encoder, a not gate/inverter. The experiment is divided into two checkpoints:

1. **Checkpoint 1** – Design the circuit with active low output.
2. **Checkpoint 2** – Convert the final answer to active high output.

Checkpoint 1: Design the circuit with active low output.

Block Diagram:



Setup:

- Place the decoder IC on the left and encoder IC on the right.
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Decoder (74138)	Encoder (74148)
● Input using decoder - ○ Switch 7 → Pin 3 (C) *(MSB)* ○ Switch 6 → Pin 2 (B) ○ Switch 5 → Pin 1 (A) *(LSB)* ● +5V → Pin 16, Pin 6 ● GND → Pin 8, Pin 4, Pin 5 *(Here, Pin 4,5,6 are enable pins of decoder)*	● Output using encoder - ○ Pin 6 (A2) → Bulb2 *(MSB)* ○ Pin 7 (A1) → Bulb1 ○ Pin 9 (A0) → Bulb0 *(LSB)* ● +5V → Pin 16 ● GND → Pin 8, Pin 5 *(Here, Pin 5 is the enable pin of encoder)*
Decoder's Pin 15 will be connected with **Encoder's Pin 12** **Decoder's Pin 14** will be connected with **Encoder's Pin 13** **Decoder's Pin 13** will be connected with **Encoder's Pin 1** **Decoder's Pin 12** will be connected with **Encoder's Pin 2** **Decoder's Pin 11** will be connected with **Encoder's Pin 3** **Decoder's Pin 10** will be connected with **Encoder's Pin 4** **Decoder's Pin 9** will be connected with **Encoder's Pin 10** **Decoder's Pin 7** will be connected with **Encoder's Pin 11**	

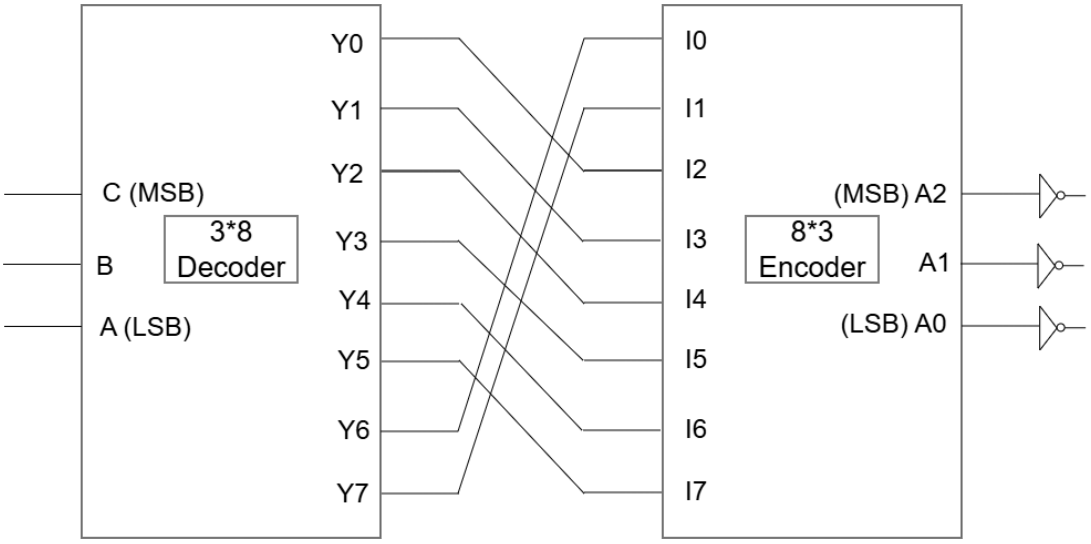
Testing the circuit:

Inputs			Actual Outputs			**Active Low Outputs (Final)**		
C (MSB)	B	A (LSB)	A_2 (MSB)	A_1	A_0 (LSB)	A_2 (MSB)	A_1	A_0 (LSB)
0	0	0	0	1	0	1	0	1
0	0	1	0	1	1	1	0	0
0	1	0	1	0	0	0	1	1
0	1	1	1	0	1	0	1	0
1	0	0	1	1	0	0	0	1
1	0	1	1	1	1	0	0	0
1	1	0	0	0	0	1	1	1
1	1	1	0	0	1	1	1	0

Checkpoint 1 - Completed.

Checkpoint 2 – Convert the final output from Checkpoint1 to active high output.

Block Diagram:



Setup:

- Everything will be the same from Checkpoint 1 except the encoder's output.
- Connect not gates with each output from the encoder to get the active high outputs.
- Place 7404(Not gate IC) beside the encoder IC.
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Not Gate (7404)
● +5V → Pin 14 ● GND → Pin 7 ● Pin 2 → Bulb 2 (MSB) ● Pin 4 → Bulb 1 ● Pin 6 → Bulb 0 (LSB)
● **Encoder's Pin 6** (A2) will be connected with **Not gate's Pin 1** ● **Encoder's Pin 7** (A1) will be connected with **Not gate's Pin 3** ● **Encoder's Pin 9** (A0) will be connected with **Not gate's Pin 5**

Testing the circuit:

Inputs			Actual Outputs		
C (MSB)	B	A (LSB)	A_2 (MSB)	A_1	A_0 (LSB)
0	0	0	0	1	0
0	0	1	0	1	1
0	1	0	1	0	0
0	1	1	1	0	1
1	0	0	1	1	0
1	0	1	1	1	1
1	1	0	0	0	0
1	1	1	0	0	1

Checkpoint 2 - Completed.